



มหาวิทยาลัยขอนแก่น

วิทยา จวิทยา ปัญญา

KHON KAEN UNIVERSITY



Website Design Best Practices

Assoc. Prof. Dr. Kanda Runapongsa Saikaew
Computer Engineering Department
Khon Kaen University



คณะวิศวกรรมศาสตร์ มหาวิทยาลัยขอนแก่น

FACULTY OF ENGINEERING KHON KAEN UNIVERSITY

13 website design best practices

1. Minimize text
2. Show, don't tell
3. Use short sentences
4. Try short paragraphs
5. Make your CTA clear and obvious
6. Reinforce actions with familiarity
7. Simplify the navigation



13 website design best practices

- 8. Optimize your design for mobile devices
- 9. Prioritize SEO
- 10. Monitor your page loading speed
- 11. Choose a color scheme that fits your branding strategy
- 12. Continuously run A/B tests
- 13. Use bullets



1. Minimize text

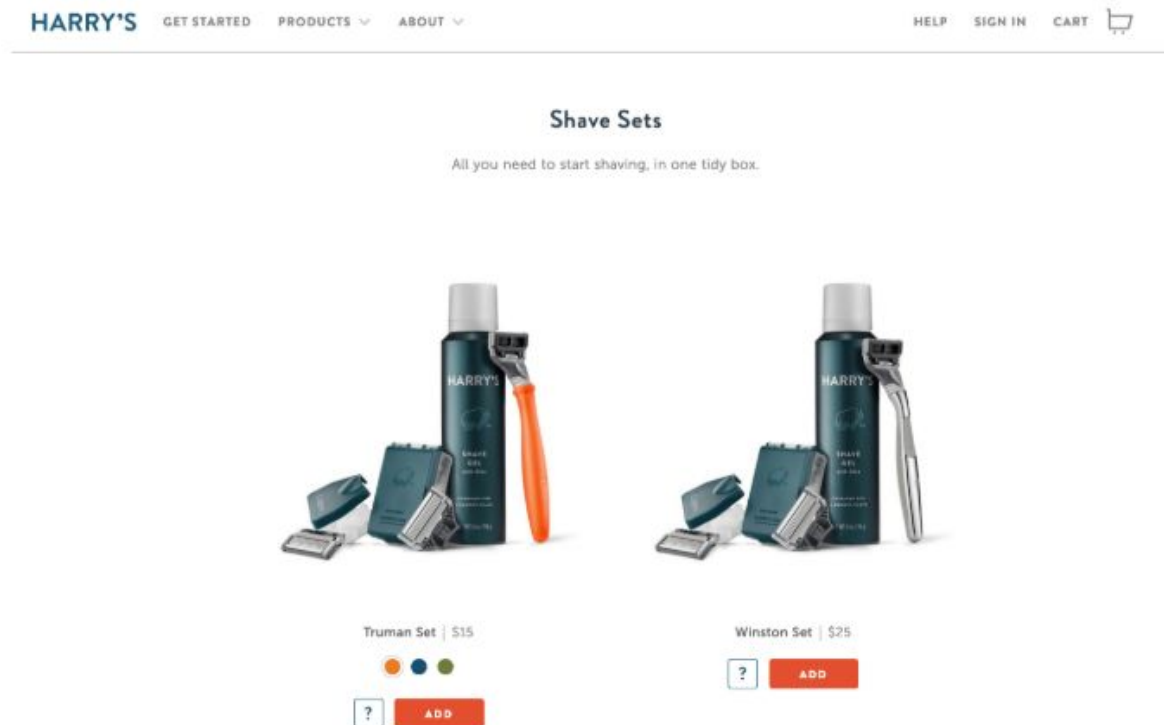
The screenshot shows the Yahoo! homepage with a dense layout of links and advertisements. At the top, there are icons for Personalize, Finance, Shop, Email, Messenger, and Help. Below these are links for Yahoo! Travel (Reserve: Flights, Cars, Hotels, Cruises, Vacations) and Get Deals: Delta Fare Sale. A search bar is present with a 'Search' button and links to 'advanced search' and 'most popular'. The left sidebar contains sections for 'New!' (Sign up for Fantasy Football - Play Toki Toki Boom), 'Shop Find Connect' (Auctions, Autos, Classifieds, Real Estate, Shopping, Travel, Careers, Maps, People Search, Personals, Yellow Pages, Chat, GeoCities, Greetings, Groups, Mail, Messenger, Mobile), 'Organize' (Addresses, Briefcase, Calendar, My Yahoo!, PayDirect, Photos, Games, Horoscopes, Kids, Movies, Music, TV, Finance, News, Sports, Weather), and 'More Yahoo!...'. The 'Yahoo! Shopping' section features a search bar and links to Computers, Electronics, Apparel, and More... It also lists 'Great Computer & Electronic Deals!' including Palm Showcase, Compaq Showcase, Sony Showcase, Last day to save on Notebooks, and Mobile Phones. The 'Web Site Directory' section is organized by subject, with categories like Business & Economy, Regional, Computers & Internet, and Society & Culture. The 'In The News' section shows a headline about an Israel blast. The 'Shopping' section features a Polaroid Digital Camera for \$69.99 and a Bed & Bath Blowout.

The screenshot shows the Google homepage with a clean and minimalist design. The address bar displays 'https://www.google.com'. The Google logo is prominently displayed in the center. Below the logo is a large search bar with a microphone icon on the right. At the bottom, there are two buttons: 'Google Search' and 'I'm Feeling Lucky'. The text 'Google offered in: ไทย' is visible at the bottom right.



2. Show, don't tell

- Instead of explaining things to your website visitors, you can simply show them.



3. Use short sentences

- Short sentences are easier to read
- If text are all over the screen, users won't know where to start reading and won't be able to digest your content
- If you need a long sentence, follow it with a short one



Sample of variety of short text

← → ↻ A Medium Corporation [US] | https://medium.com

Medium

HOME ONEZERO ELEMENTAL GEN ZORA FORGE HUMAN PARTS STARTUPS SELF TECH HEATED MODUS MORE


How to Feel Normal When the World Says You're Too Big
There are concrete ways to help your brain normalize your body (and pretty much everyone else's)
Virgie Tovar in Forge
Jul 26 · 4 min read ★


Science, Society, and So-Called Socialist Plants
Matthew MacDonald in OneZero
Jul 30 · 5 min read ★


Trump's Bigotry Is Fuel for Megan Rapinoe's Success
Donte' Stallworth in GEN
Jul 26 · 4 min read ★


Guns Kill Loved Ones but Offer Little Self-Defense
Robert Roy Britt in GEN
Jul 30 · 5 min read ★


When Everyday Sounds Drive You Crazy
What it's like to live with misophonia, a ...
Matthew Oglesby in Elemental
Jul 30 · 9 min read ★

[SEE ALL FEATURED >](#)

FROM YOUR NETWORK

Introvert กับ Extravert
หลายปีก่อน ผมมี โอกาสได้ pair programming กับเพื่อนร่วมงานสุดหล่อชื่อ เจน
Chokchai Phatharamalai in odds.team
Jul 18 · 2 min read



New from your network

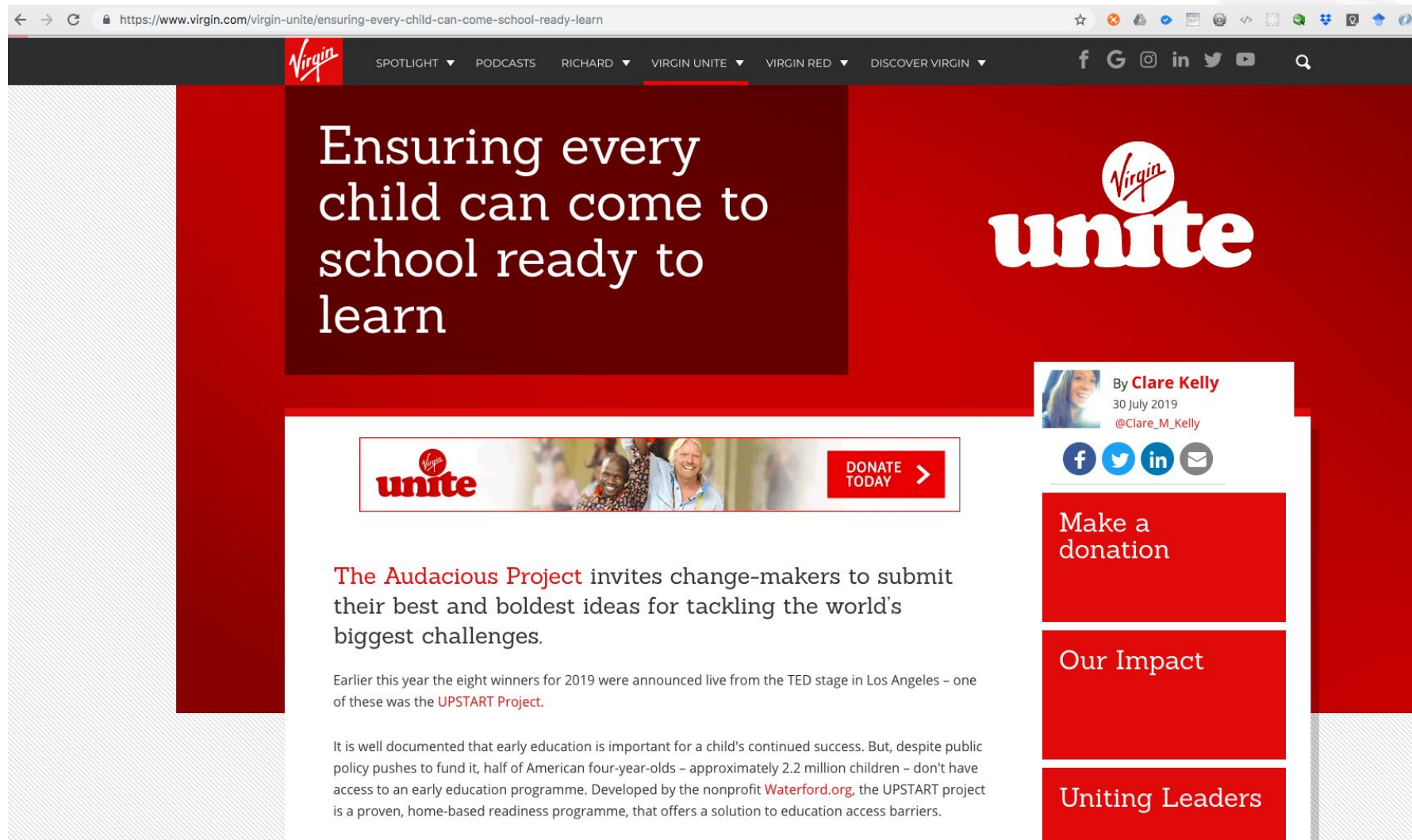



4. Try shorter paragraphs

- Use paragraph breaks to your advantage
- It's okay to write longer paragraphs, but I like to keep my homepage paragraphs to a few sentences
- Start each paragraph with the new information
 - If someone is scrolling they can quickly tell if they need to read that paragraph.



Sample website with short paragraphs

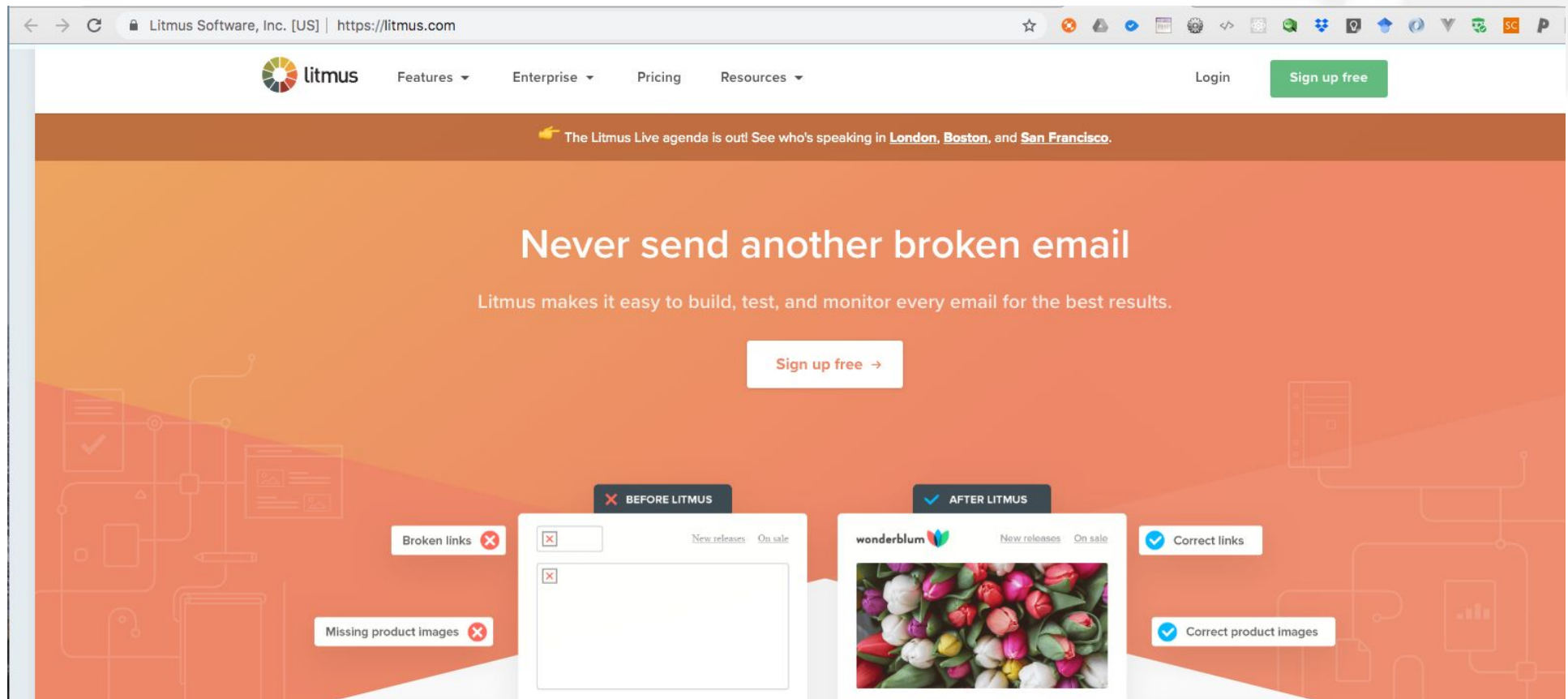


5. Make your CTA clear and obvious

- Call to Action (CTA)s should not be buried.
- They need to be big, bold, powerful, and clearly stand out as the action your visitors should take next
- Only 47% of websites have a CTA button that can be spotted in less than three seconds
- 72% of businesses don't have a CTA on their interior pages



Sample obvious and clear CTA



6. Reinforce actions with familiarity

- If your message is the same, your CTA should be the same. On every single page
- Think about how people navigate on your website
- Even if you set up a certain flow, not everyone will land on a page and convert in seconds.



Sample website with reinforce actions

Apple WATCH SERIES 4

More powerful, more colorful.

[Learn more >](#) [Buy >](#)



MacBook Pro

More power. More performance. More pro.

[Learn more >](#) [Buy >](#)

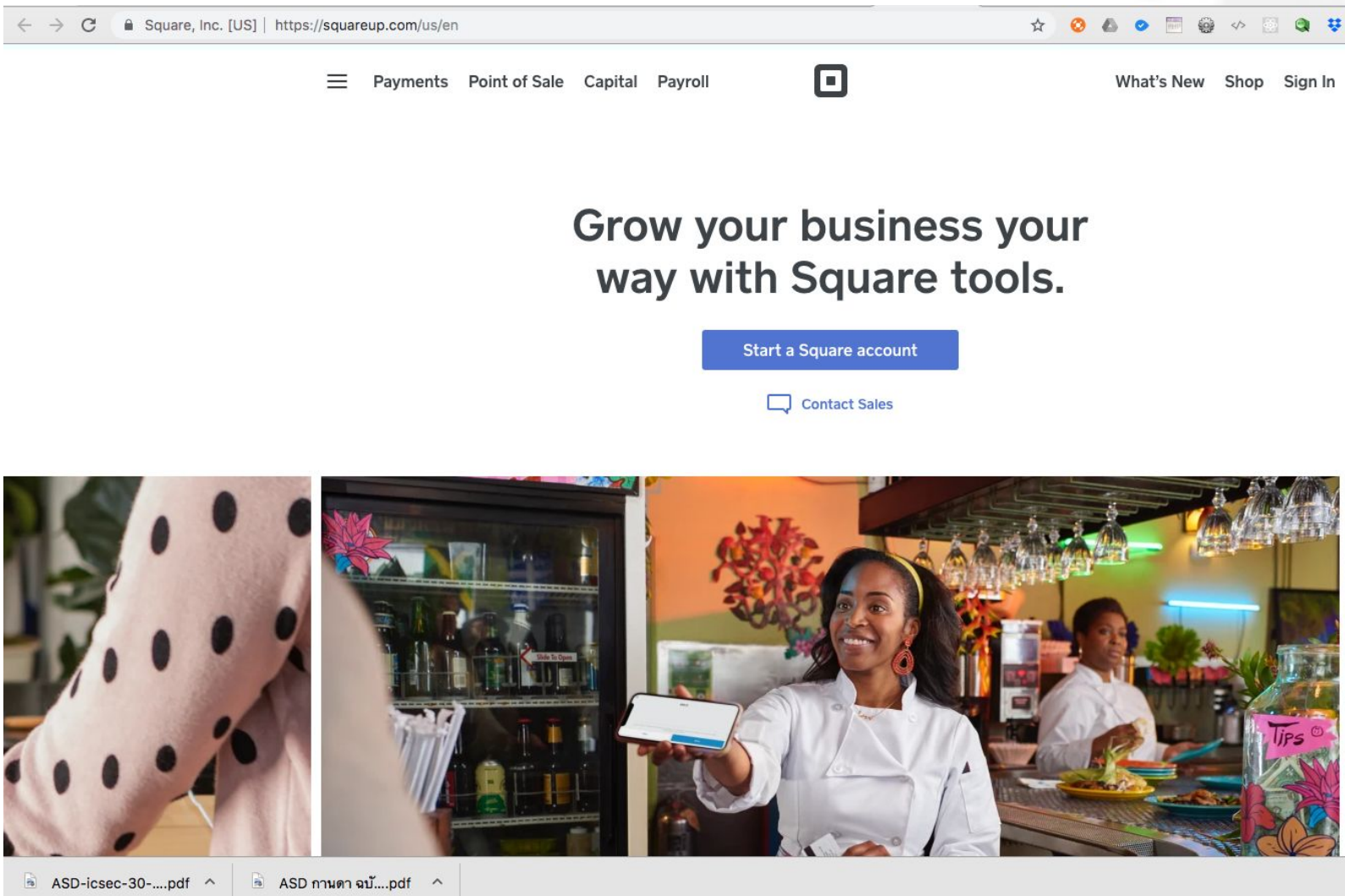


7. Simplify the navigation

- 88% of websites have the main navigation menu positioned horizontally at the top of every page
- If your menu is somewhere else, it's going to confuse your visitors
- Find ways to eliminate unnecessary menu options
 - Instead of having a “home” button, just use the website logo to link back to the home screen

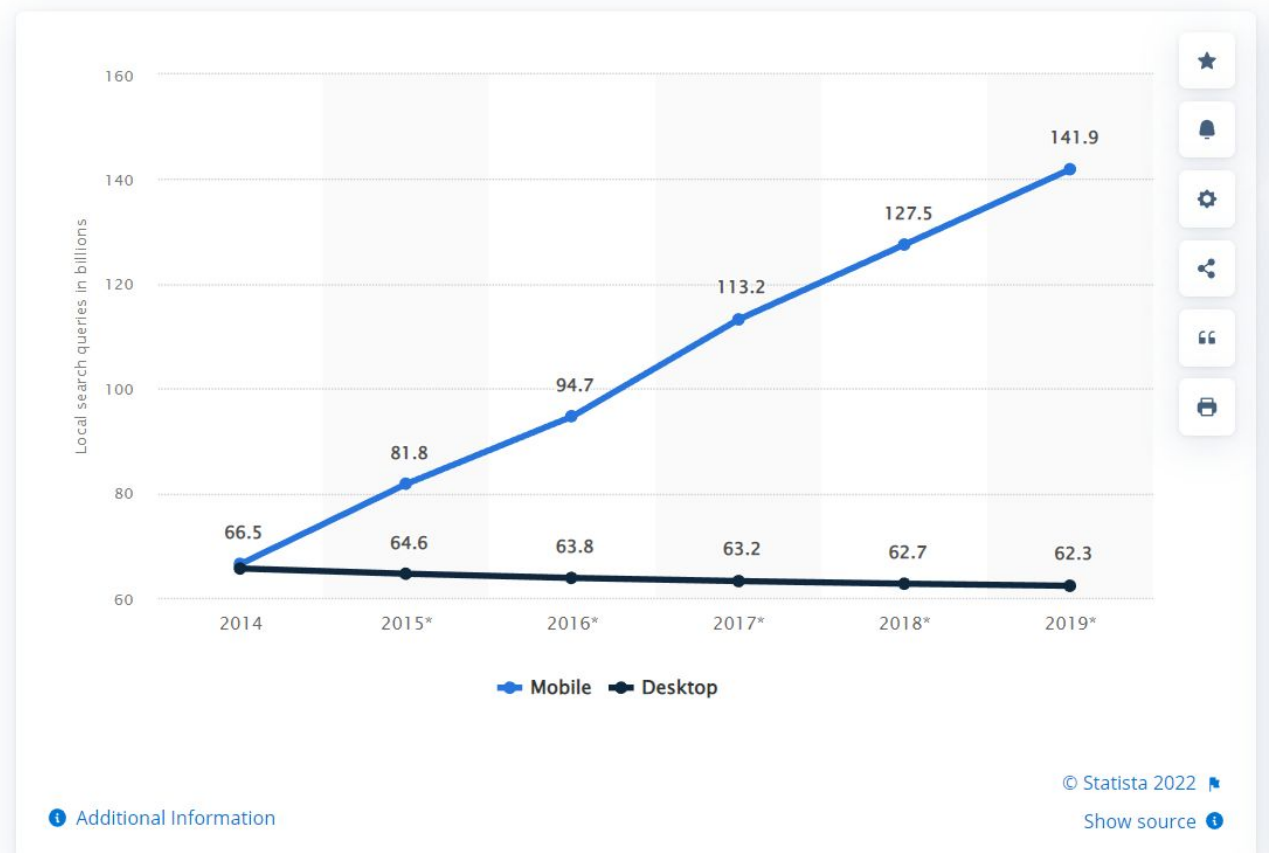


Sample website with menu at the top



8. Optimize your design for mobile devices

- Local search query volume in the United States from 2014 to 2019, by platform (*in billions*)



What are the Differences of These Terms?

- Mobile-friendly
- Mobile-optimization
- Responsive Design for Websites



Mobile-Friendly Websites

- Google's search rankings emphasize and **reward mobile-friendly websites**
- **Mobile-friendly is the bare minimum** mobile design strategy you should have for your mobile visitors
- If you aren't sure if your website is mobile-friendly, then check-out Google's **webmaster tool**
- **Mobile-friendly websites will work for mobile users, but these websites were designed for the desktop users.**



Mobile-Optimized WebSites

- All mobile-optimized websites are mobile-friendly, but not all mobile-friendly websites are mobile-optimized
- Mobile-optimization targets mobile users; these websites are designed for smaller screens
- **A mobile-optimized website will reformat itself for mobile users**
- It is not just a shrunken down version of the desktop website



Features of a Mobile-Optimized Website (1/2)

- Single column layout
- Easy, simple navigation that is “thumb friendly”
- Large graphics with white-space borders for those of us with large or clumsy fingers
- Formatted content for maximum readability



Features of a Mobile-Optimized Website (2/2)

- Limited to no need for typing
- Image file sizes are smaller for the mobile version of the website than those used for the desktop version of the website (This allows for quicker load times)
- A fewer number of features overall (minimalist, uncluttered design)



Websites Created with Responsive Design

- Responsively designed websites do exactly as phrase suggest
- They “respond” to the screen size of the device being used
- In a sense, responsive design picks up where mobile-optimization leaves off
- **Responsive design reformats and restructures websites for *any* device—regardless of screen size.**



Responsive design

- 94% of people will judge a website based on a responsive design
- A mobile-optimized website is not only essential for the user experience, but it's important for SEO purposes as well
- Google knows that 87% of smartphone owners use their devices to run an Internet search at least once per day
- The result: 70% of the first page results on Google are optimized for mobile devices.



Mobile-Friendly Test

- [https://search.google.com/test/mobile-friendly](https://search.google.com/test/mobile-friendly?id=jThrY_C4mo2KLGa0rpHs8Q)

The screenshot shows the Google Mobile-Friendly Test interface. The browser address bar displays the URL https://search.google.com/test/mobile-friendly?id=jThrY_C4mo2KLGa0rpHs8Q. The page title is "Mobile-Friendly Test" and the URL being tested is <http://gear.kku.ac.th/>.

Test results

Page loading issues [VIEW DETAILS](#)

Tested on: Jul 31, 2019 at 10:46 AM

Page is mobile friendly

This page is easy to use on a mobile device

Additional resources

- [Open site-wide mobile usability report](#)
- [Learn more about mobile-friendly pages](#)
- [Post comments or questions to our discussion group](#)

Rendered page

HTML [SCREENSHOT](#)

The rendered page shows a mobile view of the website. The header includes a "MENU" button and the text "วิศวกรรมคอมพิวเตอร์ คณะวิศวกรรมศาสตร์ มหาวิทยาลัยขอนแก่น". The main content area displays the date "27 กรกฎาคม 2562" and the text "DEEP LEARNING FOR COMPUTER VISION". Below this is a poster for "Deep Learning for Computer Vision" with the dates "13-16 สิงหาคม 2562" and the text "คณะวิศวกรรมศาสตร์ มหาวิทยาลัยขอนแก่น". The poster also features a QR code and the text "Downloaded from Software Agent, Thailand (20,000 baht)".



Exercise1 (1/2)

- Comment **at least three items** of what you think the web **should be improved based on web design best-practices**
 - For the student ID that ends with 0, 5
 - Comment on the web <https://ukraine.craigslist.org/>
 - For the student ID that ends with 1, 6
 - Comment on the web <https://blinkee.com/>
 - For the student ID that ends with 2, 7
 - Comment on the web of <https://www.domperignon.com/age-gate>
 - For the student ID that ends with 3, 8
 - Comment on the web of <https://www.paradisewater.com/orangecounty/bottled-water-huntington-beach.html>
 - For the students ID that ends with 4, 9
 - Comment on the web of <https://www.lingscars.com/>



Exercise1 (2/2)

- **Capture the part of the commented website** in the document so that the explanation can be understood more clearly
 - For example, please don't just say "bad colors" but should say
 - the colors are not well contrasted
 - the background color is not well contrasted with the text color which makes the text difficult to read.



9. Prioritize SEO

- 93% of all Internet experiences start with a search engine
- With so many people clicking on organic search listings, it's vital that you use search engine optimization (SEO) to help your site rank higher in the search results
- By investing in creating an SEO-friendly website, you will rank better in the search results and drive more traffic to your page



What is an SEO-friendly Site? (1/3)

1. Unique Title and Descriptions for all Pages

- For an SEO-friendly website, all its web pages should have their own unique meta titles and descriptions

2. Clean URL Structure

- The URL should be all lower case, descriptive, separated by dashes, and possibly contain keywords organically for more visibility.

3. Fast Loading Webpages

- No one wants to open a webpage that is too slow to load
- Image optimization can help you in increasing the page loading speed



What is an SEO-friendly Site? (2/3)

4. Unique Content

- Search engines flag websites with plagiarized content and this can significantly reduce your website's SEO friendliness

5. Create a Proper Content Structure

The general structure of your webpage should have these elements:

- Header
- H1 Tag or Page Title
- Content with Subheadings (H1, H2, H3... tags for well-structured content distribution)
- Author Bio/Information, Sharing Options
- Footer



What is an SEO-friendly Site? (3/3)

6. Responsive Design

- Responsive design adapts your website to all types of devices and fits in based on user searches

7. Internal Linking

- Internal links or hyperlinks point to different pages of the same website and help search engines and users understand how information is organised on your website and the connection between the information hierarchy and the content itself.



10. Monitor your page loading speed

- 40% of people abandon websites that take more than three seconds to load
- How can you apply this to your web design?
 - Reduce the file sizes of your images
 - Take advantage of browser caching tools
 - Reduce HTTP request
 - Improve your TTFB (time to first byte)
 - Minify and combine your files



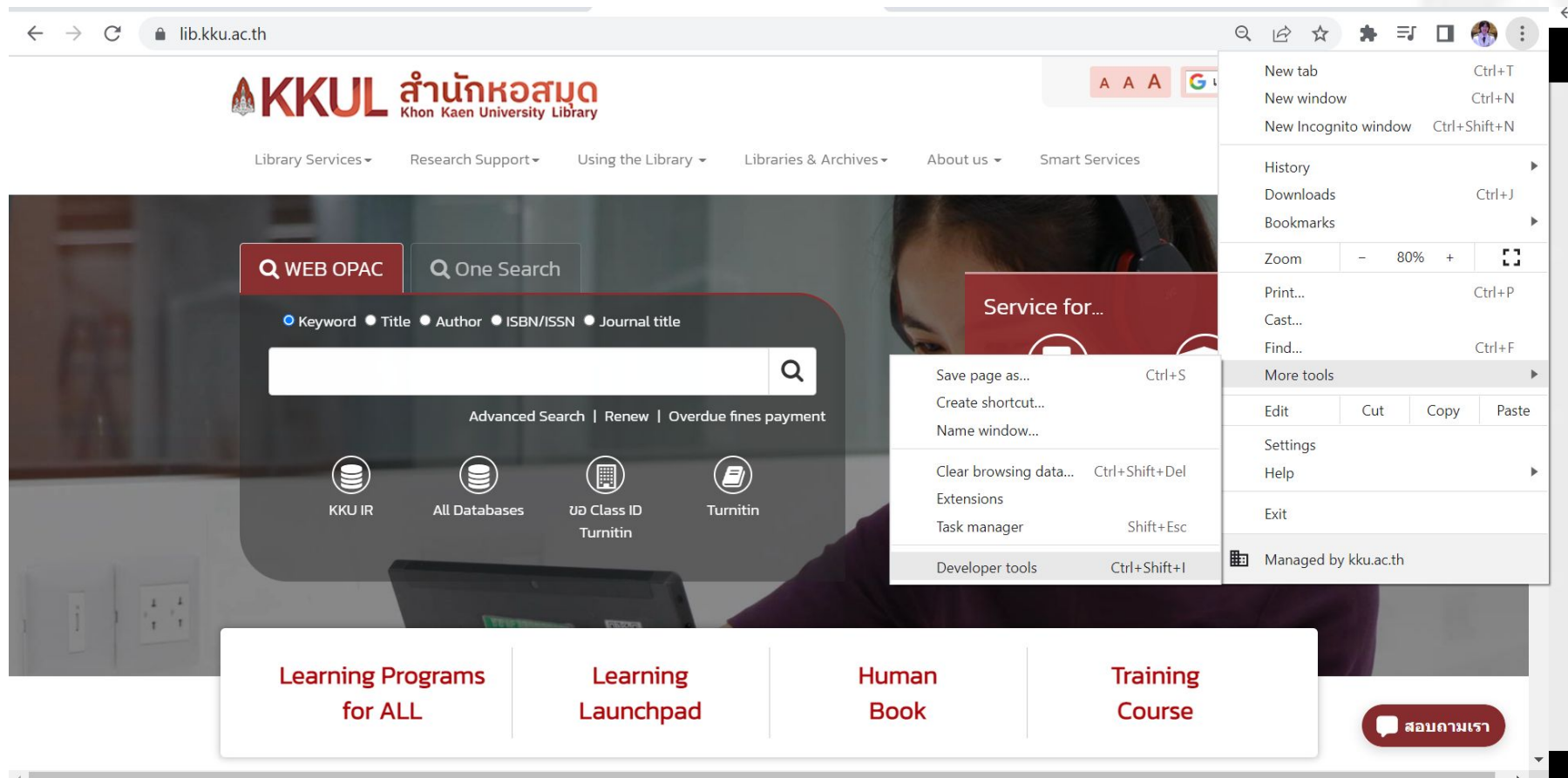
Tools to monitor loading speed

- If you use WordPress, check out the WordPress Rocket Plugin as a resource
 - <https://wp-rocket.me/blog/reduce-http-requests-speed-wordpress-site/>
- Use the Page Speed Insights tool from Google to help you monitor your loading times
 - <https://developers.google.com/speed/pagespeed/insights/>



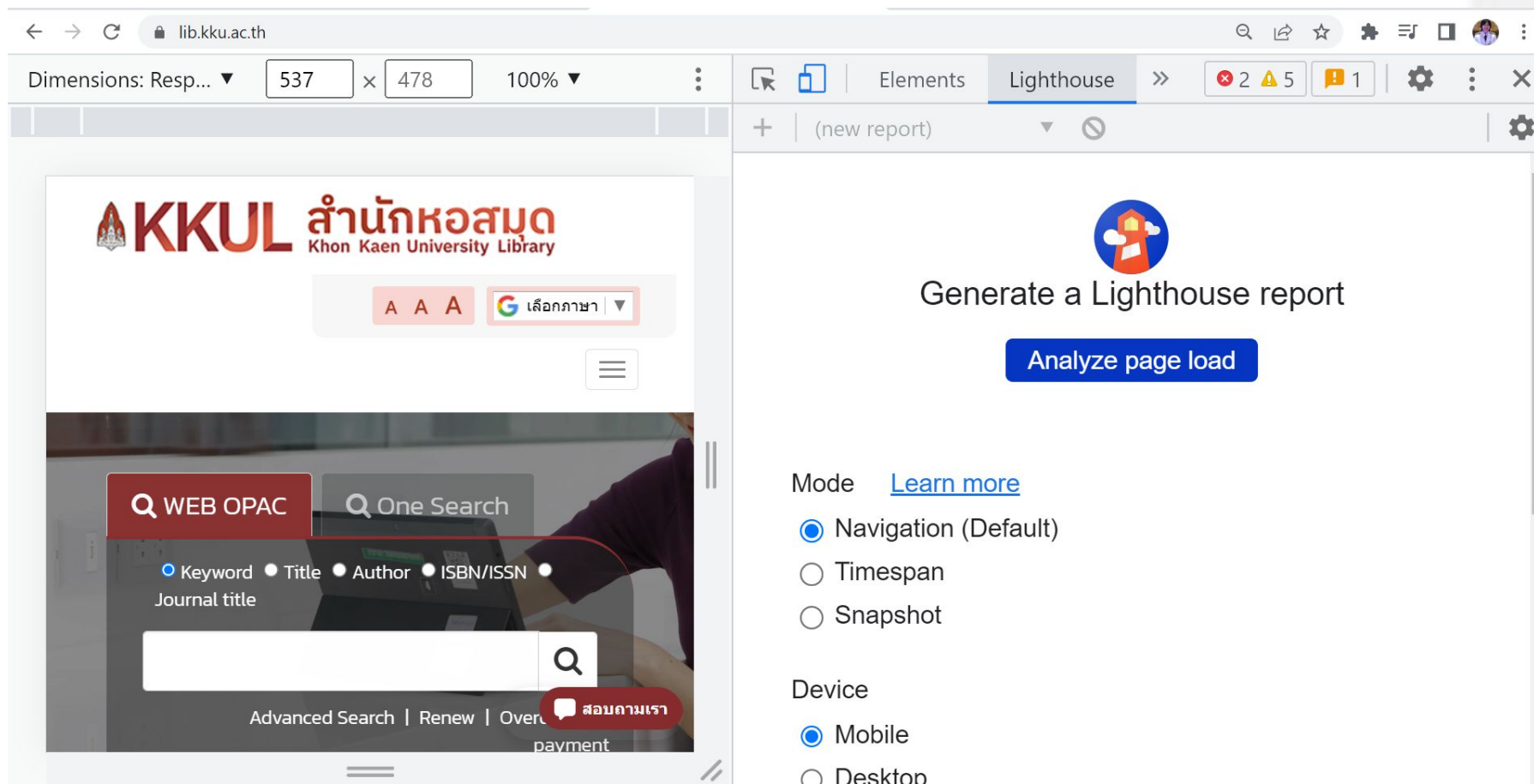
Using Chrome LightHouse (1/5)

- Click at the menu “Developer tools”



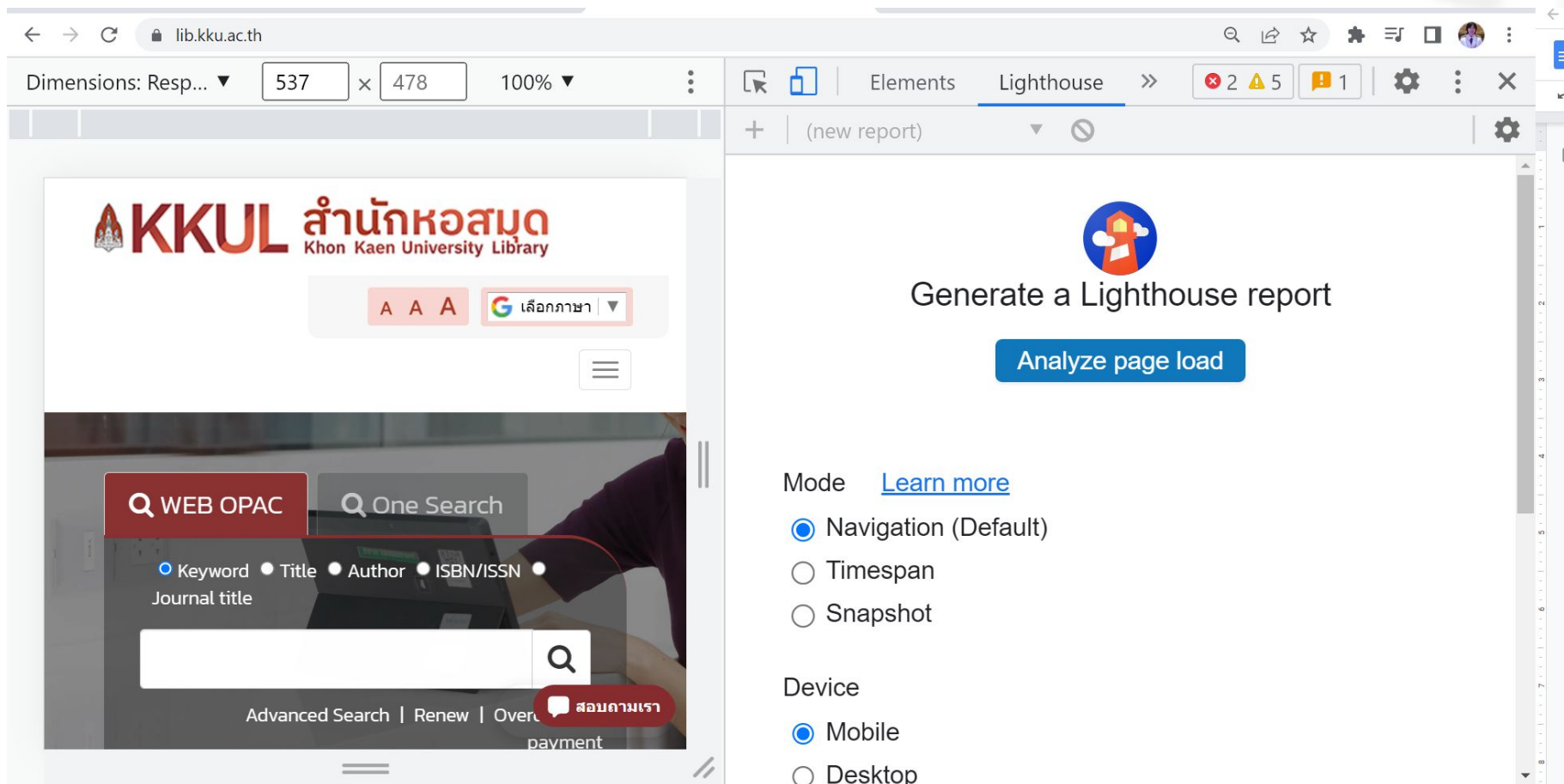
Using Chrome LightHouse (2/5)

- Click at the tab LightHouse



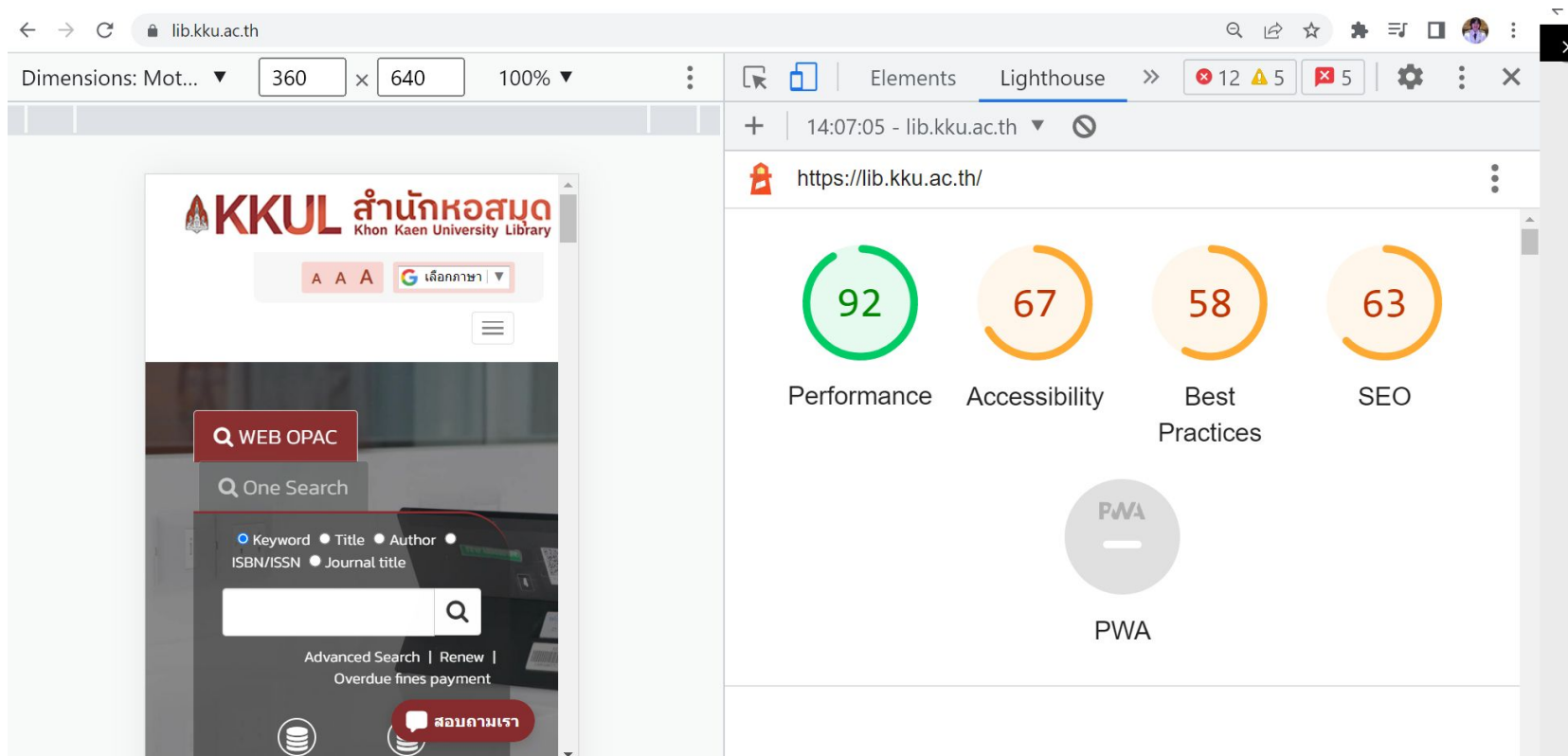
Using Chrome LightHouse (3/5)

- Click at the menu Analyze page load



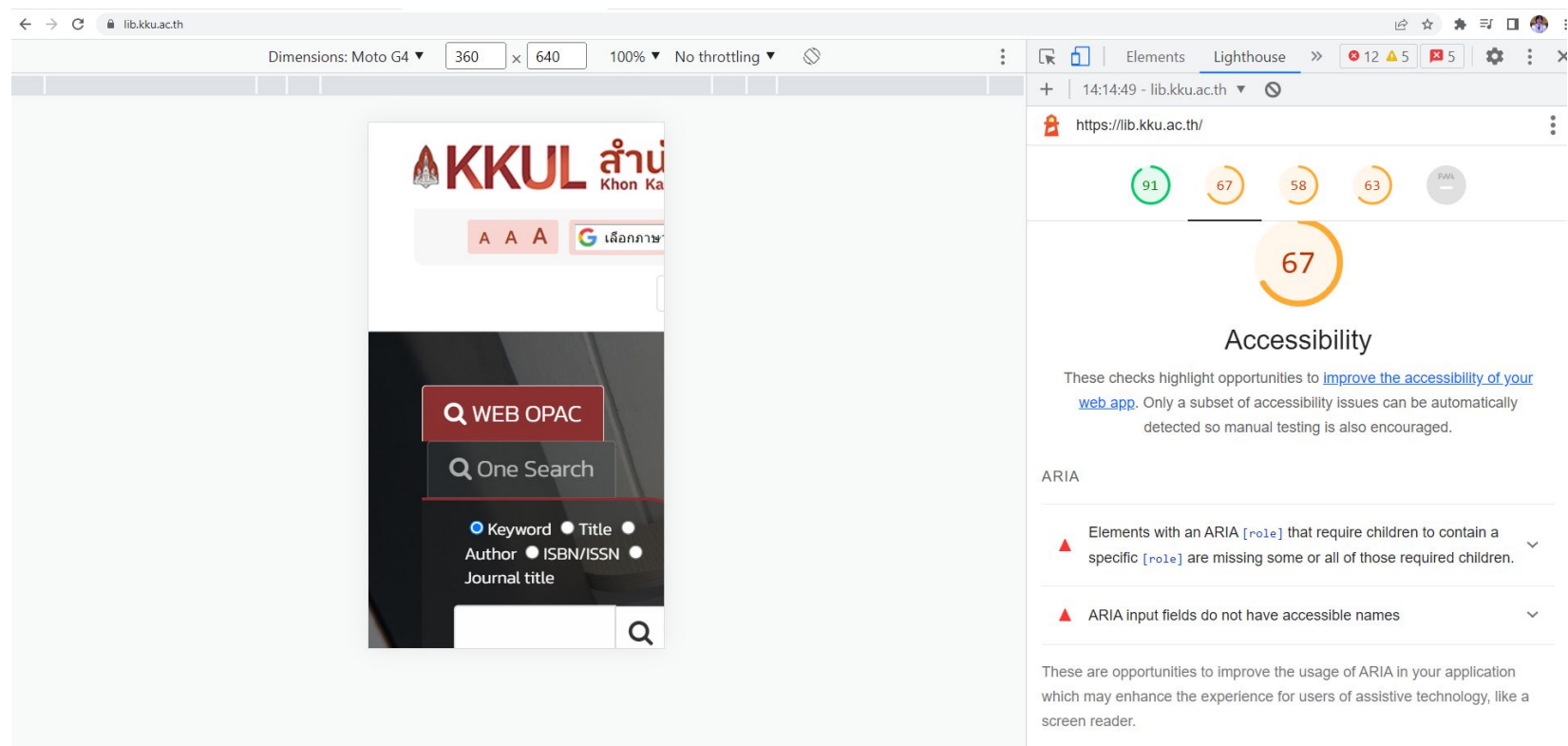
Using Chrome LightHouse (4/5)

- The report about Performance, Accessibility, Best Practices, SEO, and PWA is generated and displayed



Using Chrome LightHouse (5/5)

- The report explains what each topic measures and how you can improve your webapp

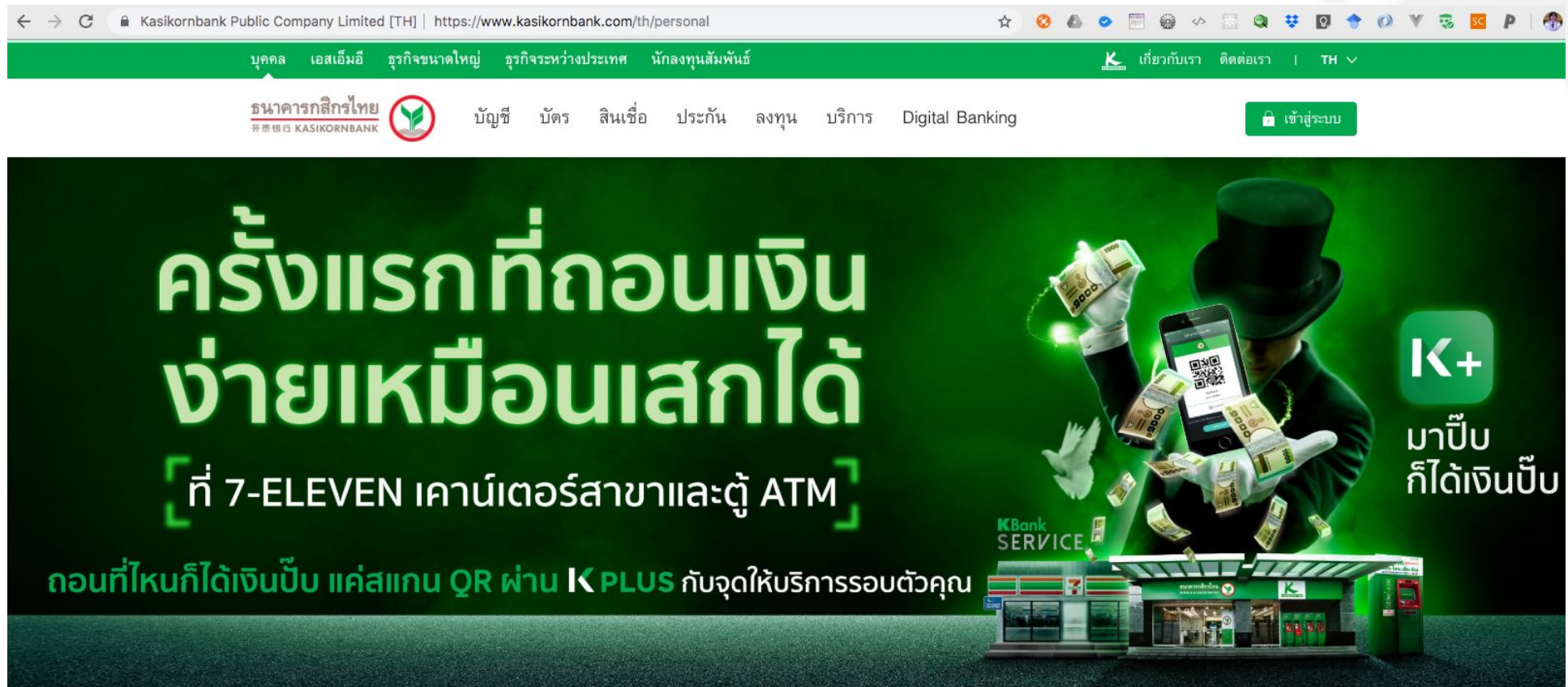


11. Choose a color scheme that fits your branding strategy

- The color choices you make on your website are more important than you think
- Your website can be considered as a product
 - 90% of that judgement is based solely on colors
- The best way to choose your website color scheme is with branding
 - Colors can increase brand recognition



Sample website with a color scheme that fits your branding strategy



12. Continuously run A/B tests

- When it comes to your website design, you can't just see it and forget it
- No website is perfect
- There are always ways for you to improve your design
- That's why you need to run A/B tests



Beginners Guide to A/B Testing

- There are two main ways to make money from your website
 - The first is to drive more traffic
 - The second is to convert more visitors into customers
- A/B testing is an effective method to figure out what changes you need to make to your website to achieve your goal



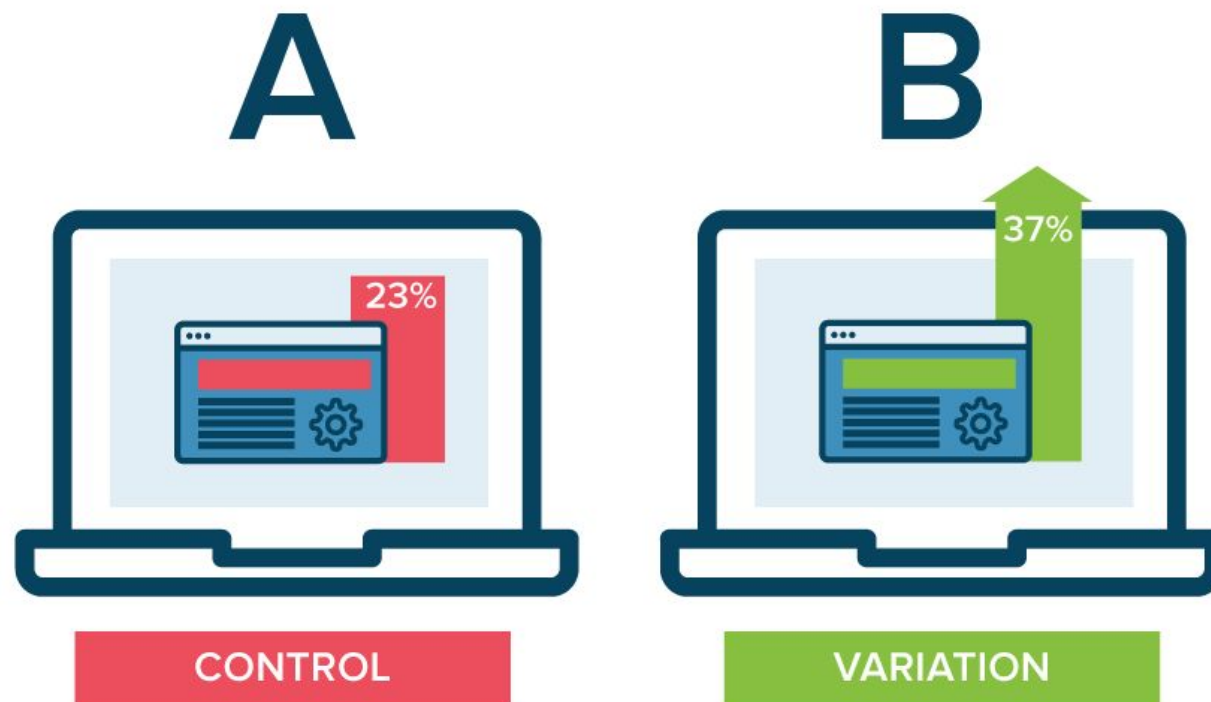
How A/B testing works

- You start by creating a hypothesis about a certain element and then run a test to see if your theory was right
- You create two different versions of your website
 - Half of your site traffic will get sent to one version
 - The other half will get directed to the variation

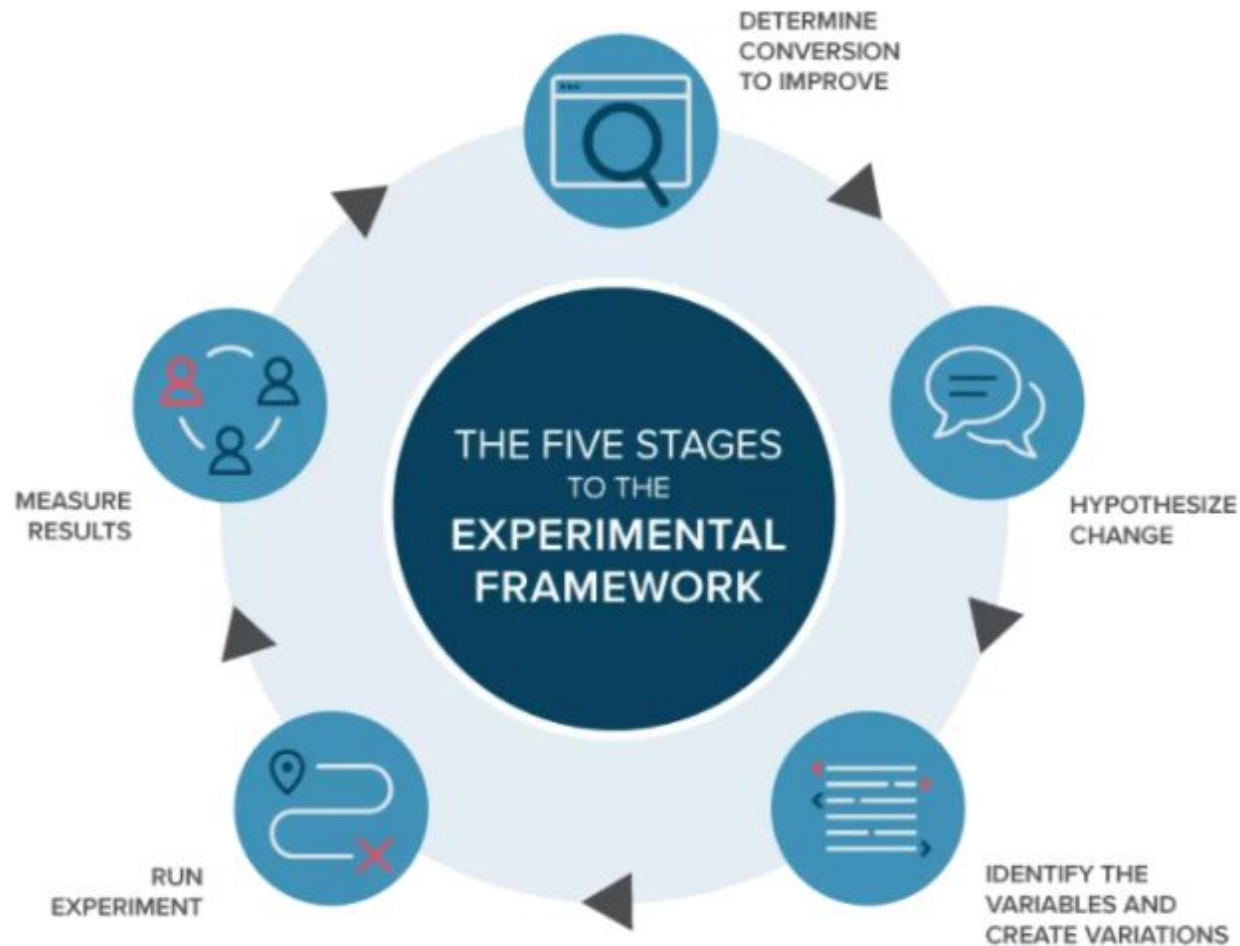


What is A/B testing?

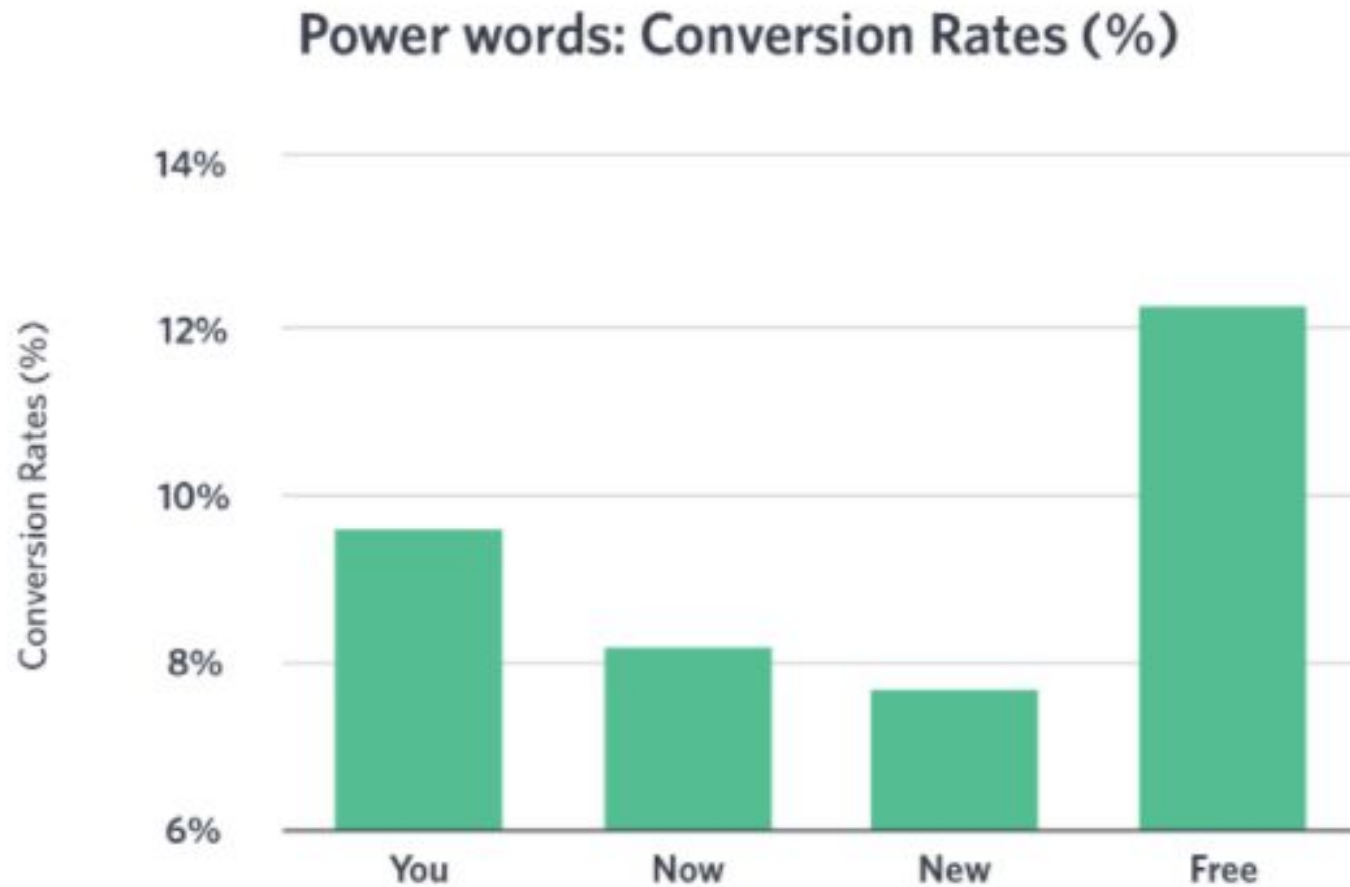
- A/B testing (also known as split testing)



Set a clear goal for your testing procedure



Test your CTA buttons



Evaluate your headings

- The test yielded conclusive results. Website visitors who saw version A filled out the form at a 27.76% higher rate than the ones who saw version B.

VERSION A vs. VERSION B

VERSION A

MarketingProfs

CREATE SUCCESSFUL SOCIAL MEDIA CAMPAIGNS FAST with SmartTOOLS – your interactive, online guide for high-ROI social media campaigns.

Your social media campaigns will win with SmartTools.

SmartTools Social Media Marketing will help you:

- Form an effective entry strategy based on market insights and core company goals
- Prepare your team and help integrate it by establishing a corporate policy, best practices, and self-learning programs
- Change and fine tune with an engaging process that requires your target audience to connect
- Spur the performance of stakeholders and focus for valuable word of mouth
- Add much more

Join today **FREE**

Access Now

Try it now step-by-step to create marketing campaigns. Join MarketingProfs now to access the SmartTools Social Media Marketing, also thousands of the world's best marketing resources. Already a member? Log in and access SmartTools here >

Our mission: We will reach out to you and ensure the value you will reap is all yours. [Join us today!](#)

VERSION B

MarketingProfs

SMART TOOLS Join today and get access to SmartTools Social Media Marketing. Your Blueprint for Social Media marketing success.

Create winning social media campaigns with SmartTools.

SmartTools Social Media Marketing will help you:

- Form an effective entry strategy based on market insights and core company goals
- Prepare your team and help integrate it by establishing a corporate policy, best practices, and self-learning programs
- Change and fine tune with an engaging process that requires your target audience to connect
- Spur the performance of stakeholders and focus for valuable word of mouth
- Add much more

Join today **FREE**

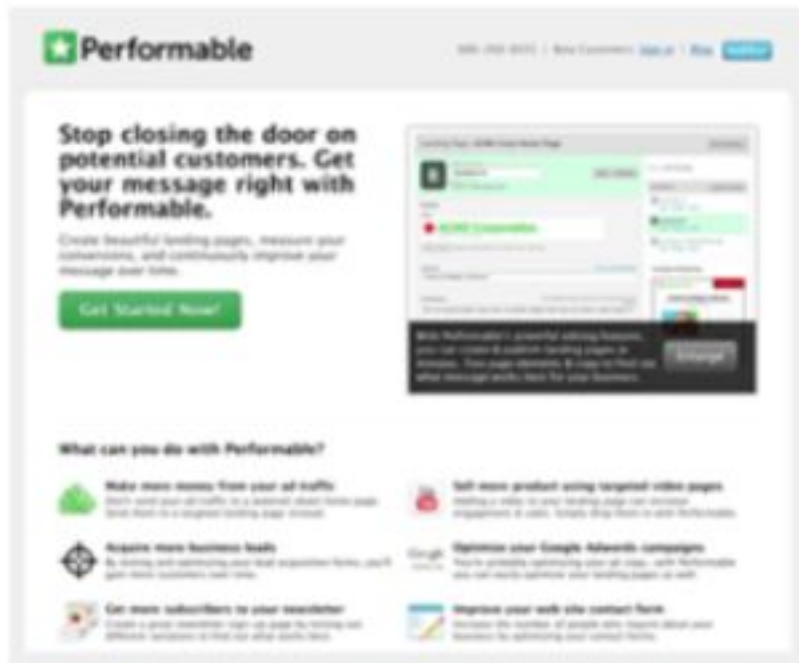
Access Now

Try it now step-by-step to create marketing campaigns. Join MarketingProfs now to access the SmartTools Social Media Marketing, also thousands of the world's best marketing resources. Already a member? Log in and access SmartTools here >

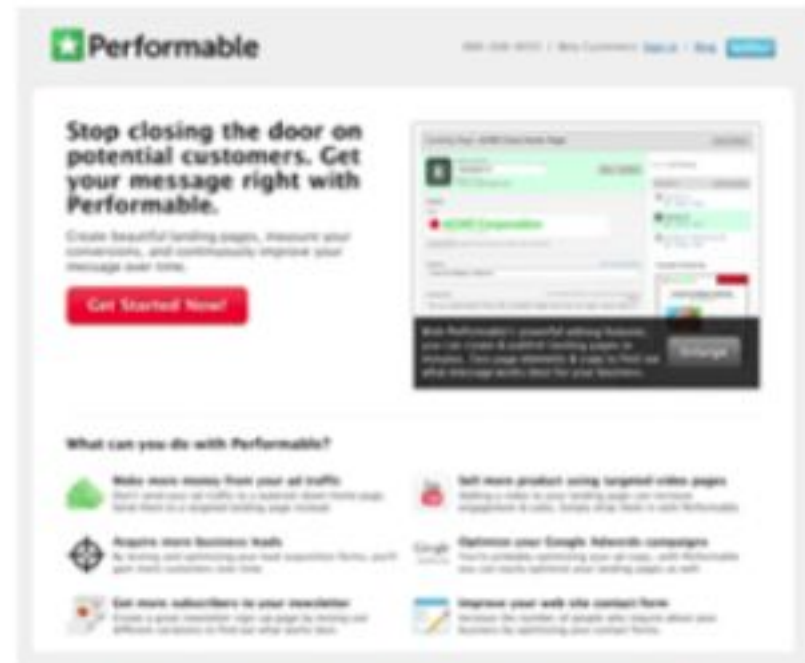
Our mission: We will reach out to you and ensure the value you will reap is all yours. [Join us today!](#)



Test different color schemes



Get Started Now!



Get Started Now!



Sum up the process

- The process
 - Set a goal
 - Come up with a hypothesis (what elements should I change to try to achieve this goal?)
 - Run an A/B test
 - Analyze the results
- If you change too many components of your website, it will be extremely difficult to accurately test your hypothesis



Select a platform to run your A/B tests

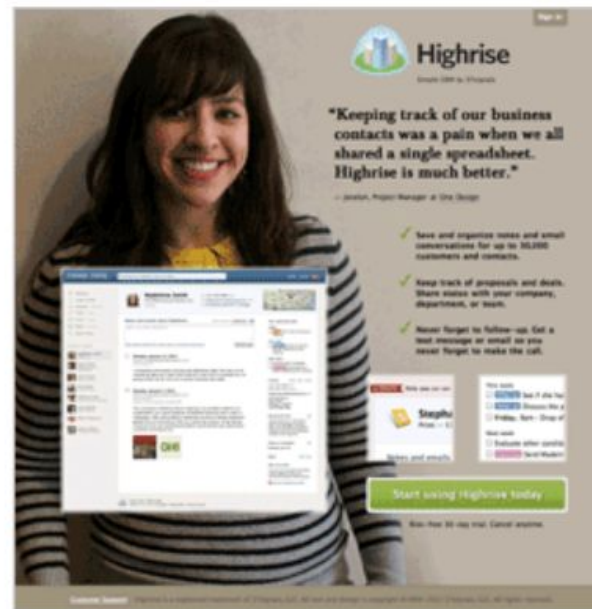
- Google Optimize
(<https://optimize.google.com/optimize/home/#/accounts>)
- A/B Testing and conversion Optimization Platform
(<https://vwo.com/>)



Examples of A/B Testing (1/6)



ORIGINAL DESIGN



PERSON DESIGN

102.5% ↑



Examples of A/B Testing (2/6)



PERSON DESIGN
102.5% ↑



LONG FORM PERSON DESIGN
22.72% ↓



คณะวิศวกรรมศาสตร์ มหาวิทยาลัยขอนแก่น

FACULTY OF ENGINEERING KHON KAEN UNIVERSITY

Examples of A/B Testing (3/6)



JOCELYN



MICHAEL

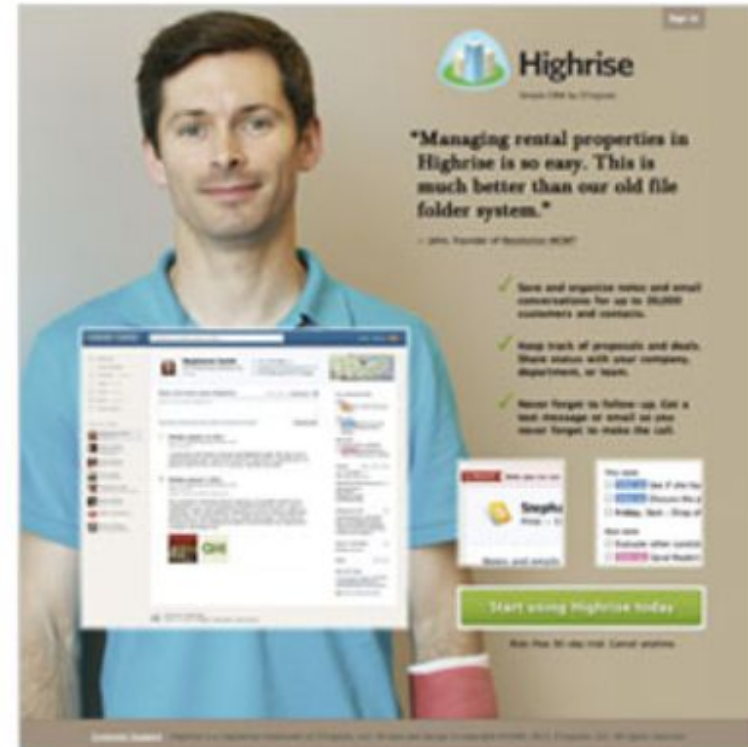
4.78% 



Examples of A/B Testing (4/6)



WILL
3.49% ↑



JOHN
3.38% ↓



Examples of A/B Testing (5/6)



MARI

0.8% ↓



BRIAN

2.2% ↑



Examples of A/B Testing (6/6)

Conclusions

- Big photos of smiling customers work
- A specific person didn't quite matter among the set of people we tested



13. Use bullets

- Instead of adding paragraphs and long-form writing, consider using lists
- In those lists, use bullet points
- Studies show that more people will look at the lists with bullet points than other formats



Why a Bulleted List?

- Bulleted lists are a great option when you need to break up a gray wall of text
- A bulleted list can:
 - Draw attention to important information
 - Support your reader's tendency to scan text
 - Communicate ideas more efficiently



Using Shapes Instead of the Dot

- Bullets don't have to be a small round dot
- Your software may offer other bullet shapes that you can easily define instead of the dot

Our current hotel special packages

- ✓ Last-minute escape
- ✓ Weekend romance
- ✓ Family movie night
- ✓ Spa retreat



Mixed Icons

Our current hotel special packages

 Last-minute escape

 Weekend romance

 Family movie night

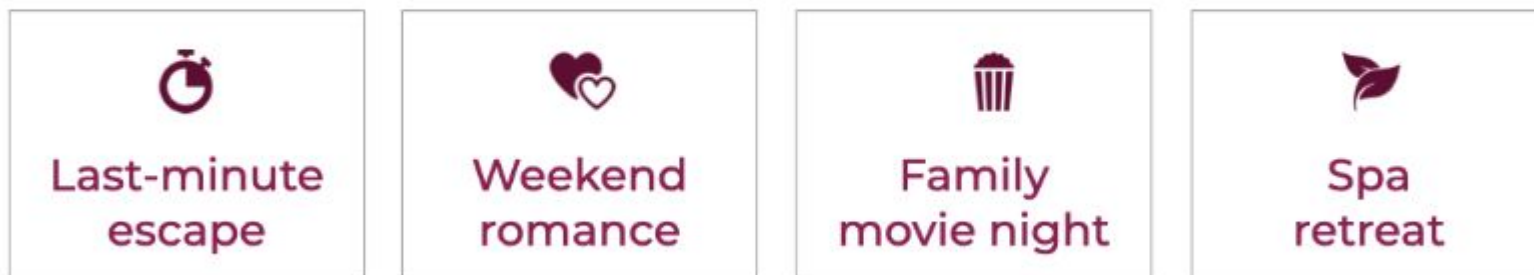
 Spa retreat



Boxes

- Sometimes your bulleted list doesn't need to look like a bulleted list
- When your list items are succinct, like simple noun phrases, a series of horizontal boxes might be an excellent alternative to a vertical list.

Our current hotel special packages



Exercise 2 (1/2)

Explain at **least three reasons why these websites have good design**

- For the students who have ID that ends with 4, 5
 - Comment on the web <https://unseen.co/>
- For the students who have ID that ends with 3, 7
 - Comment on the web <https://www.flyhyer.com/>
- For the students who have ID that ends with 2, 8
 - Comments on the web <https://www.behance.net/>
- For the students who have ID that ends with 1, 6
 - Comments on the web <https://thestandard.co/>
- For the students who have ID that ends with 0, 9
 - Comments on the web [web<https://election66.thaipbs.or.th/result>](https://election66.thaipbs.or.th/result)



Exercise2 (2/2)

- **Capture the part of the commented website** in the document so that the explanation can be understood more clearly
 - Don't just say "interesting color theme"
 - should also explain how the color theme is interesting, i.e., is it well contrasted? Does it make some parts outstanding?
 - Don't just say "easy to read"
 - should explain how it is easy to read, i.e., is the font size large enough? Does the spacing between lines have proper width



References (1/2)

- <https://www.quicksprout.com/website-design-best-practices/>
- <https://www.quicksprout.com/beginners-guide-ab-testing/>
- <https://www.tytonmedia.com/blog/51-insane-web-design-statistics-2016/>
- <https://signalvnoise.com/posts/2991-behind-the-scenes-ab-testing-part-3-final>



References (2/2)

- <https://www.statista.com/statistics/434152/local-mobile-desktop-search-query-volume-usa/>
- <http://torspark.com/mobile-friendly-vs-mobile-optimized-vs-responsive-design/>

